

Hilbert–Schmidt Operators with Increasing Standard-Compression Spectrum

Run fa_banach_001, agent_lane_14

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Status and source question

This is a `candidate_partial_likely_valid` result for Roman Drnovšek, *Triangularizability of trace-class operators with increasing spectrum*, arXiv:1508.07766 [1]. Question 3 on PDF page 3 asks:

Suppose that K is a compact operator on $L^2(X, \mu)$ that has increasing spectrum relative to standard compressions. Does K admit a standard triangularization (which need not be multiplicity-free)? In particular, is K decomposable?

Question 3. *Suppose that K is a compact operator on $L^2(X, \mu)$ that has increasing spectrum relative to standard compressions. Does K admit a standard triangularization (which need not be multiplicity-free)? In particular, is K decomposable?*

It is worth mentioning that an affirmative answer to Question 3 would extend the well-known de Pagter's theorem (see [5] or [1] Theorem 9.19]) asserting that a positive compact quasinilpotent operator K on $L^2(X, \mu)$ is decomposable. Namely, it is easy to show (see, e.g., the beginning of [3] page 3520]) that positivity of K implies that the operator PKP is quasinilpotent for each standard projection P , so that K has increasing spectrum in this case.

In this paper we consider Question 3 in the case of trace-class kernel operators. In Section 2 the underlying measure space is the unit interval $[0, 1]$, while in Section 3 we add atoms to it, and we prove the main result of the paper (Theorem 7). In Section 4 we slightly improve Theorem 2 that is the main result of [3]. In the rest of this section we recall some relevant definitions and facts.

An operator K on $L^2(X, \mu)$ is called a *kernel operator* if there exists a measurable function $k : X \times X \rightarrow \mathbb{C}$ such that, for every $f \in L^2(X, \mu)$, the equality

$$(Kf)(x) = \int_X k(x, y)f(y)d\mu(y)$$

holds for almost every $x \in X$. The function k is called the *kernel* of the operator K . The kernel operator K is positive if and only if its kernel k is nonnegative almost everywhere. If the kernel operator K with kernel k has the modulus $|K|$, then the kernel of $|K|$ is equal to $|k|$ almost everywhere. For more details on kernel operators we refer the reader to [7].

Given a compact operator T on $L^2(X, \mu)$, let $\{s_j(T)\}_j$ be a decreasing sequence of singular values of T , i.e., the square roots of the eigenvalues of the self-adjoint operator T^*T , where T^* denotes the adjoint of T . If $\sum_j s_j(T) < \infty$, the operator T is said to be a *trace-class operator*. In this case, the *trace of T* is defined by

$$\text{tr}(T) = \sum_{n=1}^{\infty} \langle Tf_n, f_n \rangle,$$

where $\{f_n\}_{n=1}^{\infty}$ is any orthonormal basis of $L^2(X, \mu)$. By Lidskii's Theorem, the trace of a trace-class operator T is equal to the sum of all eigenvalues of T counting algebraic multiplicity. We will use freely the equality $\text{tr}(AT) = \text{tr}(TA)$ that holds for every trace-class operator T and every operator A . For more details on trace-class operators see, e.g., [6].

For a measurable $E \subseteq [0, 1]$, write P_E for multiplication by $\mathbf{1}_E$. Increasing spectrum means

$$\sigma(P_E K|_{P_E L^2}) \subseteq \sigma(P_F K|_{P_F L^2}) \quad (E \subseteq F).$$

The source's Theorem 5 answers the interval case when K is trace class, its kernel is continuous,

and its lattice modulus is also trace class. The result below removes all three restrictions, retaining only the strictly weaker Hilbert–Schmidt hypothesis.

Main result

Theorem 1 (Hilbert–Schmidt interval case). *Let $K \in \mathcal{S}_2(L^2[0, 1])$ have increasing spectrum relative to standard compressions. Then K is quasinilpotent and admits a standard triangularization.*

More precisely, if $k \in L^2([0, 1]^2)$ is the kernel of K and T is the positive Hilbert–Schmidt operator with kernel $|k|$, then T is quasinilpotent. A standard triangularizing chain supplied for T by the ideal-triangularization form of de Pagter’s theorem consists of K -invariant standard subspaces as well.

Proof intuition

The trace of K need not exist, and a measurable kernel has no meaningful diagonal values. Both apparent losses are avoided by starting with the second power. Every Hilbert–Schmidt compression has trace-class powers from exponent two onward, and increasing spectrum makes all of their power traces zero.

For n disjoint localization sets, take an alternating sum over all their unions. Boolean inclusion–exclusion cancels every block word that misses a set; a word of length n that visits all n sets is necessarily a permutation. Lebesgue differentiation therefore says that the sum of all oriented n -cycle products is zero almost everywhere. A shortest nonzero cycle cannot cancel in this sum: all its backward chords except the closing edge would create shorter cycles, so only its n rotations remain.

Thus the absolute kernel $|k|$ has zero cyclic moments of every order at least two. A separate spectral lemma shows that an Hilbert–Schmidt operator with all those power traces zero must be quasinilpotent. De Pagter’s theorem then provides the standard chain, and $|Kf| \leq T|f|$ transfers its invariance to K . The full compact problem remains open because a compact operator need not possess either an L^2 kernel or any trace-class power.

Three lemmas

We first isolate the only spectral fact needed at the end.

Lemma 2 (vanishing power traces). *If $A \in \mathcal{S}_2$ and*

$$\mathrm{tr}(A^n) = 0 \quad (n \geq 2),$$

then A is quasinilpotent.

Proof. Let (λ_j) be the nonzero eigenvalues of A , repeated according to algebraic multiplicity. Weyl’s inequality gives $\sum_j |\lambda_j|^2 < \infty$, and Lidskii’s theorem applied to $A^n \in \mathcal{S}_1$ gives

$$0 = \mathrm{tr}(A^n) = \sum_j \lambda_j^n \quad (n \geq 2).$$

Suppose that $r = r(A) > 0$, and list the finitely many peripheral eigenvalues as $\lambda_1, \dots, \lambda_m$. Simultaneous recurrence on the compact torus gives integers $n_q \rightarrow \infty$ such that $(\lambda_j/r)^{n_q} \rightarrow 1$ for every $j \leq m$.

The remaining eigenvalues have modulus at most some $\rho < r$ outside an ℓ_2 tail of modulus at most $r/2$. Consequently

$$r^{-n_q} \sum_{j>m} |\lambda_j|^{n_q} \longrightarrow 0 :$$

the finite part is $O((\rho/r)^{n_q})$, while the tail is at most $r^{-2}2^{-(n_q-2)} \sum_j |\lambda_j|^2$. Dividing the zero power sum by r^{n_q} and passing to the limit therefore gives $0 = m$, a contradiction. Thus $r(A) = 0$. $\square$

For $n \geq 2$, define the cyclic product

$$c_n(x_1, \dots, x_n) = k(x_1, x_2)k(x_2, x_3) \cdots k(x_n, x_1)$$

and its complete symmetrization

$$F_n(x_1, \dots, x_n) = \sum_{\pi \in S_n} c_n(x_{\pi(1)}, \dots, x_{\pi(n)}).$$

Products of Hilbert–Schmidt kernels around a finite cycle are integrable; this follows first for finite simple kernels from the trace formula and then for L^2 kernels by the Schatten Hölder bound. Hence $F_n \in L^1$.

Lemma 3 (localized inclusion–exclusion). *Under the hypotheses of Theorem 1,*

$$F_n = 0 \quad \text{almost everywhere on } [0, 1]^n \quad (n \geq 2).$$

Proof. Drnovšek’s Theorem 4 [1] uses only compactness and the compressions $P_{[0,t]}$ to show that K is quasinilpotent. For every measurable E , increasing spectrum then gives

$$\sigma(P_E K P_E|_{P_E L^2}) \subseteq \sigma(K) = \{0\}.$$

Thus $K_E = P_E K P_E$ is quasinilpotent. Since $K_E \in \mathcal{S}_2$, Lidskii gives

$$\text{tr}(K_E^n) = 0 \quad (n \geq 2). \tag{1}$$

Fix pairwise disjoint measurable sets $E_1, \dots, E_n$. For $J \subseteq \{1, \dots, n\}$, put $P_J = P_{\cup_{j \in J} E_j}$. Taking the Boolean alternating sum of (1) yields

$$0 = \sum_{J \subseteq [n]} (-1)^{n-|J|} \text{tr}((P_J K P_J)^n). \tag{2}$$

Expand each trace into block words. If a word uses precisely the index set S , its coefficient in (2) is

$$\sum_{J \supseteq S} (-1)^{n-|J|} = \begin{cases} 1, & S = [n], \\ 0, & S \neq [n]. \end{cases}$$

A surviving word has length n and uses every one of the n indices, so its index word is a permutation. The standard trace formula for cyclic products of Hilbert–Schmidt kernel operators therefore turns (2) into

$$\int_{E_1 \times \cdots \times E_n} F_n(x_1, \dots, x_n) dx_1 \cdots dx_n = 0.$$

Take the E_i to be equal-radius intervals centered at distinct points x_i . They are disjoint for all sufficiently small radii. The ordinary Lebesgue differentiation theorem on $[0, 1]^n$, applied to the integrable function F_n , gives $F_n(x_1, \dots, x_n) = 0$ for almost every tuple with distinct coordinates. The collision diagonals have measure zero, proving the lemma. $\square$

Lemma 4 (cycle elimination). *For every $n \geq 2$, $c_n = 0$ almost everywhere.*

Proof. We induct on n . Since $F_2(x_1, x_2) = 2c_2(x_1, x_2)$, the base case follows from Lemma 3.

Assume the assertion through length $n - 1$. If c_n were nonzero on a set of positive measure, we could choose a tuple of distinct coordinates $(x_1, \dots, x_n)$ such that $c_n \neq 0$, $F_n = 0$, and every shorter cycle on every subtuple vanishes. This simultaneous choice is legitimate because only finitely many null sets and their coordinate pullbacks are excluded.

For $1 \leq i < j \leq n$ with $j - i \leq n - 2$, the forward path

$$k(x_i, x_{i+1}) \cdots k(x_{j-1}, x_j)$$

is nonzero. Closing it with $k(x_j, x_i)$ gives a cycle of length at most $n - 1$, so the induction hypothesis forces $k(x_j, x_i) = 0$. In any directed Hamiltonian cycle on the ordered vertices $1, \dots, n$, avoiding all such backward edges leaves only

$$1 \rightarrow 2 \rightarrow \cdots \rightarrow n \rightarrow 1$$

and its n cyclic rotations: after the unique allowed backward edge $n \rightarrow 1$, every other edge must move forward, and visiting all vertices forces their natural order. Hence every summand of F_n vanishes except those n rotations, all equal to $c_n(x_1, \dots, x_n)$. This gives $0 = F_n = nc_n \neq 0$, a contradiction. $\square$

Proof of the theorem

The quasinilpotence of K was already obtained in the proof of Lemma 3. Let T be the positive integral operator with kernel $|k|$. Since $|k| \in L^2([0, 1]^2)$, $T \in \mathcal{S}_2$ and is compact. For every $n \geq 2$, $T^n \in \mathcal{S}_1$, and the Hilbert–Schmidt cyclic trace formula plus Lemma 4 gives

$$\mathrm{tr}(T^n) = \int_{[0,1]^n} |k(x_1, x_2)| \cdots |k(x_n, x_1)| dx_1 \cdots dx_n = 0.$$

Lemma 2 shows that T is quasinilpotent.

The ideal-triangularization corollary of de Pagter’s theorem for positive compact quasinilpotent operators [3, 4] supplies a maximal standard chain invariant under T . If $L^2(E)$ belongs to this chain and f is supported in E , then

$$|Kf|(x) \leq \int_0^1 |k(x, y)| |f(y)| dy = (T|f|)(x).$$

The right side is supported in E , so Kf is supported in E . Thus the same standard chain is invariant under K , proving the asserted standard triangularization. $\square$

Verification, upgrade attempts, and limitations

The proof uses no numerical approximation. A finite script included as `code/check_cycle_combinatorics.py` verifies through $n = 8$ both the Boolean coefficient in Lemma 3 and that exactly the n cyclic rotations survive the forbidden-backward-edge condition. It is a regression check, not a substitute for the displayed proofs.

Four focused upgrades were attempted. First, the source’s trace-class assumption on the modulus was replaced by Hilbert–Schmidt power traces. Second, trace class of K itself was removed,

initially for continuous kernels. Third, Boolean localization removed continuity and produced the full theorem above. Fourth, the route was tested against arbitrary compact operators. That last extension stops for a structural reason: a compact operator can belong to no $\mathcal{S}_p$ with finite p , so no positive power need have a trace; it also need not possess an L^2 kernel whose absolute value defines a positive compact majorant. Norm approximation by finite-rank operators does not preserve increasing spectra or invariant standard subspaces. Consequently the source question for arbitrary compact operators remains open.

A bounded novelty search on 11 August 2026 used the exact source title and the combinations `increasing spectrum relative to standard compressions`, `Hilbert-Schmidt`, and `standard triangularization`, together with the run’s registry, solution, attempt, and proof-gap indexes. It found the original 2009 paper [2] and the 2017 source paper, but no later Hilbert–Schmidt theorem. Novelty confidence is moderate pending specialist review. Human review should focus on the L^2 cyclic trace formula, the null-set pullbacks in the induction, and the precise ideal-triangularization corollary used in the last paragraph.

References

- [1] R. Drnovšek, *Triangularizability of trace-class operators with increasing spectrum*, J. Math. Anal. Appl. 447 (2017), 1102–1115; arXiv:1508.07766.
- [2] L. W. Marcoux, M. Mastnak, and H. Radjavi, *Triangularizability of operators with increasing spectrum*, J. Funct. Anal. 257 (2009), 3517–3540.
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- [4] H. Radjavi and P. Rosenthal, *Simultaneous Triangularization*, Springer-Verlag, New York, 2000.